Applying Kimbrough (2019) to the Growth Scenario

Sensor placement for sewer capacity risk, and how its risk framework could be used in our growth simulation

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1. Purpose

This report has two parts. Part A explains what Kimbrough's dissertation does and what it found. Part B looks at how the team's growth simulation currently works and suggests how Kimbrough's risk framework could be adapted to it, so that sensors are placed to notice the growth that is most likely to cause an overflow rather than all growth equally.

Source: Kimbrough, H. R. (2019). *Optimal sensor placement for sewer capacity risk management* [Doctoral dissertation, Colorado State University]. Page numbers in brackets refer to this dissertation.

Part A: The dissertation

2. The problem it addresses

In the United States, sewers release about 850 billion gallons of untreated sewage from combined systems each year, alongside roughly 40,000 sanitary sewer overflows and 400,000 basement backups (p. ii). Many of these could be avoided if the pipes close to failure were found early enough to clean or repair. The difficulty is that a network has thousands of pipes and inspecting even a few of them is expensive, so the chance of any one pipe failing is hard to estimate (p. ii).

Kimbrough defines failure as **surcharge**: the water rises above the top of the pipe ($d/D > 1$), so the pipe has no spare capacity left (p. 11). The aim of the work is to find the small number of high-risk locations at lower cost by placing level sensors more intelligently and moving them over time (pp. 10, 23–24).

3. Method

Step	What is done
Risk score	Adapted from Failure Modes and Effects Analysis (FMEA). Each location gets a risk priority number, $R = S \times O$, where S is severity and O is occurrence (likelihood), each scored 1 to 10 (p. 250).
Occurrence	Level data from 456 manholes in seven US sewer systems. Logistic regression links the share of time a pipe runs at or above **0.7 of its depth** to the chance of surcharge, and this is converted to a 1 to 10 score (pp. 66, 83–87).
Severity	Scored 1 to 10 from aerial imagery of what surrounds each location, such as homes, roads and waterways (p. 256).
Clustering	Moran's I on a 14,600-manhole network with 250 recorded overflows shows that overflows cluster in space (pp. 99–100).
Placement	An agent-based simulation in AnyLogic on a synthetic 400-manhole network, where sensors move between manholes. Simulated annealing, sequential search, greedy search and a multi-phase strategy are compared (pp. 120–125, Chapter 5).

4. Main findings

- Thirty days of level data ranked risky locations with about 80% accuracy (pp. 83, 87, 252).
- Thirty days of five-minute readings give 8,640 data points, compared with a single snapshot from a CCTV inspection (pp. 251–252).
- Moving sensors guided by a search method found the target number of high-risk locations at about one seventh of the cost of a sequential search (p. 252).
- A multi-phase search (sample broadly, focus on the highest-severity areas, then widen) was the most robust strategy and the easiest for operators to follow (pp. 253–254).

5. Scope and limitations

- The failures studied come mainly from maintenance problems, especially blockages, which cause nearly half of overflows (p. 22). Infiltration and inflow are not the focus, and structural failure and odour are excluded (pp. 11, 22).
- Placement methods are tested only in simulation, on a synthetic network under static conditions (pp. 30, 258).
- The approach assumes failures cluster, so isolated failures may be missed (p. 30). The severity score does not consider overflow volume (p. 256).
- All data and regulations are from the US, and the author works for a flow-monitoring company (p. iv).

Part B: Application to the growth scenario

6. How the growth scenario works now

The team's Simulation 2.5 asks: if new houses connect somewhere in the catchment, which manholes would show it, and where must sensors be to notice it? The settings are held in the project's configuration file.

Element	Current setting
Growth sites	New dwellings added at one manhole at a time, out of 328 candidate manholes.
Growth sizes	10, 30, 80 and 150 dwellings, plus a stress case of 700.
Load per dwelling	2.3 people $\times$ 200 L per person per day, with a peak factor of 2.0.
Cases	Five cases for what cannot yet be measured: Dry, Design wet (infiltration 0.11 L/s per 100 m), Severe (0.40 L/s per 100 m), and two Design wet variants (age-weighted infiltration, outside inflow $\times$1.5).
Runs	328 sites $\times$ 5 sizes $\times$ 5 cases = 8,200 EPA SWMM runs. Each records the rise in water level at every manhole.
Detection rules	A float switch notices growth if the depth passes 150 mm. A level sensor notices it if the depth rises by at least 10, 25 or 50 mm.
Placement	Greedy selection: add the manhole that notices the most growth events not yet covered, for the worst case or the average case.

For example, with the 25 mm rule and the worst-case objective, six sensors notice 61% of all growth events.

The gap. Every growth event currently counts the same. Ten dwellings that raise the level in a nearly empty pipe by 25 mm count as much as 150 dwellings that push a nearly full pipe towards surcharge. Coverage measures how much growth is seen, but not whether the growth that matters most is seen.

7. What Simulation 2.6 already showed

Simulation 2.6 added Kimbrough's risk score as a separate placement goal (severity $\times$ likelihood, with likelihood based on the 0.7 d/D threshold and severity based on the number of homes upstream). With three sensors the two goals chose different places:

Plan	Share of total risk watched	Growth noticed (worst case)
Growth plan	7%	39%
Risk plan	14%	28%

The risk plan favours trunk mains with over 1,000 homes upstream, which are choke points, while the growth plan favours places where new housing shows as a level rise. Using either goal alone gives up part of the other.

8. Proposed adaptation: risk-weighted growth coverage

The idea is to keep Kimbrough's $R = S \times O$ but change what occurrence means. In his work it is the chance that an existing pipe surcharges because of maintenance problems. For the growth problem it becomes the chance that **the new housing** pushes a pipe past the 0.7 d/D threshold.

Part	Kimbrough (existing risk)	Proposed (growth risk)
Occurrence	Share of time the pipe already runs above 0.7 d/D	Whether, and by how much, a growth event pushes a pipe above 0.7 d/D
Severity	Surroundings from aerial imagery	Homes upstream or pipe size for now; land use and receiving waters when data is available
Risk	Per location	Per growth event
Sensor goal	Find failing pipes	Notice the highest-risk growth events first

Steps

1. For each growth event e (site, size and case), take the depth before and after growth at each manhole from the saved SWMM runs, and convert both to d/D using the pipe diameter.
2. Score occurrence O_e from 1 to 10 by how far the event pushes the most affected manhole towards or past 0.7 d/D. An event that moves a pipe from 0.5 to 0.8 scores high; one that moves a pipe from 0.2 to 0.25 scores low.
3. Score severity S_e from 1 to 10 at the manhole where the threshold is crossed, using the stand-in available now (homes upstream or pipe size).
4. Give each event a weight $w_e = S_e \times O_e$.
5. Replace plain coverage with weighted coverage:

$$\text{Weighted coverage} = \frac{\sum_{e \in \text{noticed}} w_e}{\sum_e w_e}$$

6. Run the same greedy placement, but choose each new sensor by the weight it adds rather than the number of events it adds.

Why this fits the industry partner's problem

- It answers the question the utility cares about: not only whether growth is seen, but whether the growth that could lead to an overflow is seen.
- It merges the growth plan and the risk plan into one goal instead of two competing ones.
- It stays within the agreed scope (overflow from growth), and does not bring blockages back in.
- It uses wording utilities already know (severity and likelihood), which helps explain the placement.
- It needs no new SWMM runs. It rescores the 8,200 saved runs, so it can be built and tested quickly.

9. Other ideas from the dissertation

Idea	Possible use
Severity tie-breaker	When two manholes notice similar growth, choose the one where an overflow would cause more harm.
Fixed plus roving sensors	A few permanent sensors at the modelled locations plus one or two moved each month, compared in the cost step using Kimbrough's finding that moving sensors cut search cost to about one seventh.
Thirty days of data	Supports a short, focused data request: about one month of normal-day level records may be enough to rank locations and to replace the assumed normal band in Simulation 2.6.

10. Limitations of the proposal

- Severity is a simple stand-in until land use, receiving water or overflow history data is available.
- The 0.7 threshold comes from US data. The utility's own design or alarm level should replace it if one is provided.
- The SWMM model is not yet calibrated, so d/D values are model estimates. Results show where risk is likely, not exact values.
- Kimbrough's occurrence model was built for existing pipes. Using the threshold for future growth is an adaptation that has not been tested in the literature.

11. Next step

Build the weighted coverage as a new version (Simulation 2.7) without changing Simulations 2.5 or 2.6, compare its sensor locations with the growth plan and the risk plan, and report the trade-off in a short table for the team.